Class room activity-1(Problem solving)

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Decision Tree Induction – Construct a Decision Tree Using ID3 Algorithm

1. Aim:

To construct a decision tree using the ID3 algorithm to classify loan applicants as Approved or Rejected based on their income, credit score, and existing loan status. The process includes dataset preparation, entropy calculation, Information Gain analysis, decision tree construction, and classification of new applicants.

2. Introduction:

Decision Tree Induction is a supervised machine learning technique used to build a tree-like model from a labelled dataset. It is widely used in classification and decision-making applications because its rules are easy to understand and interpret.

In a banking system, a decision tree can help analyze loan applications using applicant information. It divides the dataset into smaller subsets based on predictor attributes until a final class is assigned.

The ID3 algorithm, which stands for Iterative Dichotomiser 3, constructs a decision tree by selecting the attribute with the highest Information Gain at each split. It uses Entropy to measure the uncertainty in the dataset.

Important components of a decision tree:

- Root node: The first attribute selected for splitting the dataset.
- Internal node: An attribute used for making a further decision.
- Branch: A possible outcome or value of an attribute.
- Leaf node: The final classification, such as Approved or Rejected.
- Splitting: Dividing a dataset into subsets based on an attribute.

3. Problem Statement:

A financial company receives loan applications from customers with different income levels, credit scores, and existing loan commitments. Manually reviewing every application may take time and can result in inconsistent decisions.

Therefore, the company wants to construct a decision tree that learns from previous loan decisions and predicts whether a new applicant's loan should be approved or rejected.

The system uses the following three attributes:

1. Income level
2. Credit score
3. Existing loan status

The final output is a binary classification: Approved or Rejected.

4. Sample Dataset:

The following dataset contains 12 sample records. The records are illustrative and created for demonstrating the ID3 algorithm.

Applicant	Income	Credit Score	Existing Loan	Loan Decision
A1	High	Good	No	Approved
A2	High	Good	Yes	Approved
A3	High	Poor	No	Approved
A4	High	Poor	Yes	Rejected
A5	Medium	Good	No	Approved
A6	Medium	Good	Yes	Approved
A7	Medium	Poor	No	Approved
A8	Medium	Poor	Yes	Rejected
A9	Low	Good	No	Approved
A10	Low	Good	Yes	Rejected
A11	Low	Poor	No	Rejected
A12	Low	Poor	Yes	Rejected

Explanation of the dataset:

The dataset contains information about each applicant's financial characteristics and the final loan decision.

- Income represents the applicant's earning level.
- Credit Score represents the applicant's creditworthiness.
- Existing Loan indicates whether the applicant already has a loan.
- Loan Decision is the known class used to train the decision tree.

The dataset contains 7 Approved records and 5 Rejected records, giving a total of 12 records.

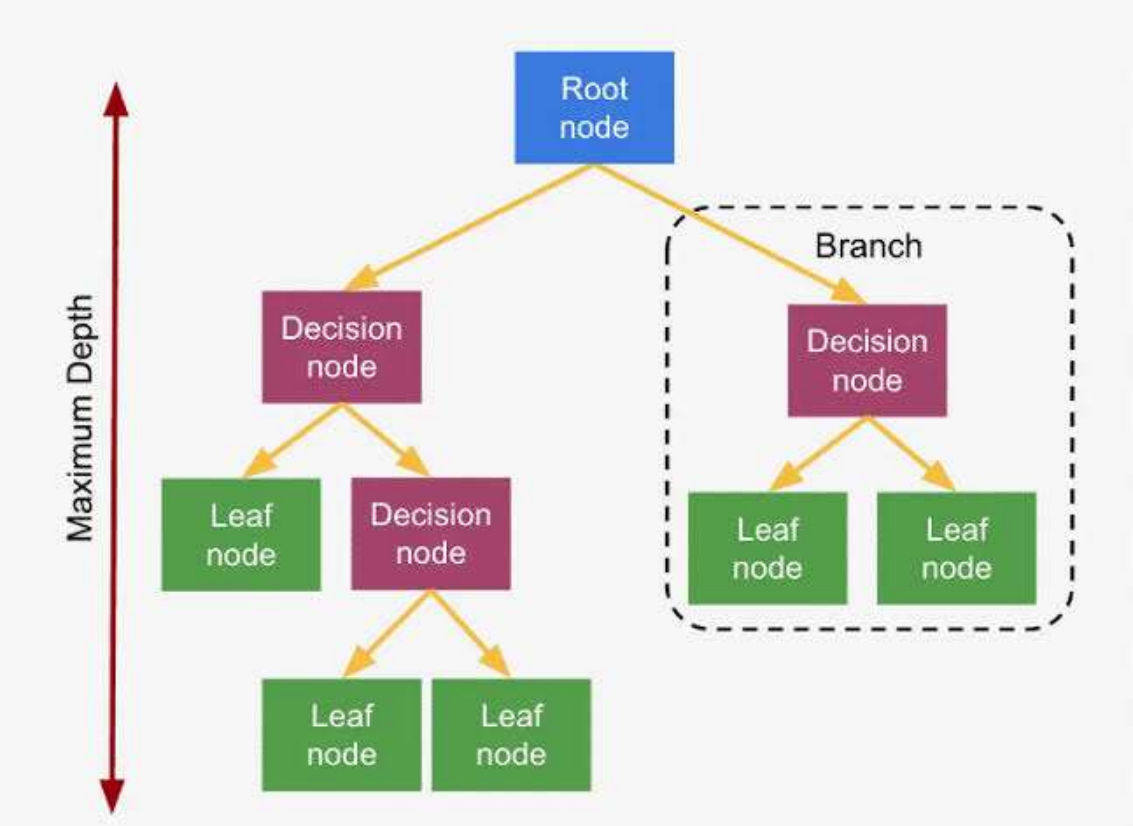
5. Target Class and Predictor Attributes:

The attributes are classified into target and predictor variables as follows.

Attribute type	Attribute name	Description
Target class	Loan Decision	Final output: Approved or Rejected
Predictor 1	Income	High, Medium, or Low
Predictor 2	Credit Score	Good or Poor
Predictor 3	Existing Loan	Yes or No

The predictor attributes are used to predict the target class. During the training process, ID3 compares the Information Gain of the predictor attributes and selects the most informative attribute for splitting.

6. Working Principle of the ID3 Algorithm:



The ID3 algorithm follows a top-down, greedy approach. It selects the best attribute at each stage and creates branches for its possible values.

Step 1: Start with the training dataset

Step 2: Calculate Entropy

Step 3: Calculate Information Gain for each attribute

Step 4: Select the attribute with the highest Information Gain

Step 5: Split the dataset into branches

Step 6: Repeat until a stopping condition is reached

Final decision tree: Approved / Rejected

The process continues until the records in a subset belong to one class, no predictor attributes remain, or a suitable stopping condition is met. If a subset cannot be separated completely, the majority class can be used as the leaf prediction.

7. Entropy Calculation:

Entropy measures the amount of uncertainty or impurity in the target class. If all records belong to the same class, entropy is zero. If the classes are evenly distributed, entropy is higher.

Entropy formula:

$$ H(S)=-p_A\log_2(p_A)-p_R\log_2(p_R) $$

Where:

- $H(S)$ = entropy of the dataset.
- p_A = proportion of Approved applications.
- p_R = proportion of Rejected applications.
- $\log_2$ = logarithm to base 2.

Step 1: Count the target classes:

Loan Decision	Number of records
Approved	7
Rejected	5
Total	12

Therefore,

$$ p_A=\frac{7}{12}=0.5833 $$
$$ p_R=\frac{5}{12}=0.4167 $$

Step 2: Substitute into the formula:

$$ H(S)=-\frac7{12}\log_2\frac7{12} -\frac5{12}\log_2\frac5{12} $$
$$ H(S)=-(0.5833)(-0.7776) -(0.4167)(-1.2630) $$
$$ H(S)=0.4536+0.5263 $$
$$ \boxed{H(S)\approx0.9799} $$

Interpretation: The original dataset has an entropy of approximately 0.9799, indicating uncertainty because it contains both Approved and Rejected records.

8. Information Gain Calculation:

Information Gain measures how much an attribute reduces the entropy of the dataset after splitting.

The formula is:

$$ IG(S,X)=H(S)-H(S|X) $$

Where:

- $IG(S,X)$ is the Information Gain of attribute X.
- $H(S)$ is the original entropy.
- $H(S|X)$ is the weighted entropy after splitting by attribute X.

The attribute with the highest Information Gain is selected for the split.

8.1 Information Gain for Income:

First, divide the dataset according to the three income values.

Income	Approved	Rejected	Total
High	3	1	4
Medium	3	1	4
Low	1	3	4

The entropy for each group is calculated separately.

High income:

$$ H(\text{High})=-\frac34\log_2\frac34 -\frac14\log_2\frac14 $$
$$ H(\text{High})=0.8113 $$

Medium income:

$$ H(\text{Medium})=0.8113 $$

Low income:

$$ H(\text{Low})=-\frac14\log_2\frac14 -\frac34\log_2\frac34 $$
$$ H(\text{Low})=0.8113 $$

Calculate the weighted entropy:

$$ H(S|\text{Income})= \frac4{12}(0.8113)+ \frac4{12}(0.8113)+ \frac4{12}(0.8113) $$
$$ H(S|\text{Income})=0.8113 $$

Therefore,

$$ IG(\text{Income})=0.9799-0.8113 $$
$$ \boxed{IG(\text{Income})=0.1686} $$

8.2 Information Gain for Credit Score:

Divide the dataset into Good and Poor credit scores.

Credit Score	Approved	Rejected	Total
Good	5	1	6
Poor	2	4	6

For Good credit score:

$$ H(\text{Good})=-\frac56\log_2\frac56 -\frac16\log_2\frac16 $$
$$ H(\text{Good})\approx0.6500 $$

For Poor credit score:

$$ H(\text{Poor})=-\frac26\log_2\frac26 -\frac46\log_2\frac46 $$
$$ H(\text{Poor})\approx0.9183 $$

Weighted entropy:

$$ H(S|\text{Credit})= \frac6{12}(0.6500)+ \frac6{12}(0.9183) $$
$$ H(S|\text{Credit})\approx0.7842 $$

Therefore,

$$ IG(\text{Credit Score})=0.9799-0.7842 $$
$$ \boxed{IG(\text{Credit Score})\approx0.1957} $$

8.3 Information Gain for Existing Loan:

Divide the dataset into applicants who have an existing loan and those who do not.

Existing Loan	Approved	Rejected	Total
No	5	1	6
Yes	2	4	6

The class distributions are identical to those obtained for Credit Score. Hence:

$$ H(S|\text{Existing Loan})\approx0.7842 $$

$$ IG(\text{Existing Loan})=0.9799-0.7842 $$
$$ \boxed{IG(\text{Existing Loan})\approx0.1957} $$

8.4 Comparison of Information Gain:

Attribute	Weighted Entropy	Information Gain
Income	0.8113	0.1686
Credit Score	0.7842	0.1957
Existing Loan	0.7842	0.1957

Decision: Credit Score and Existing Loan have equal Information Gain, approximately 0.1957. We select Credit Score as the root node to break the tie. Either is acceptable if the tie-breaking rule is stated and followed consistently.

9. Attribute Selection for the Next Splits:

After selecting Credit Score as the root, ID3 repeats the Information Gain calculation on each subset. It is important to select the highest-gain attribute again at every new node.

9.1 For Credit Score = Good

This subset contains 6 records: 5 Approved and 1 Rejected.

$$ H(S_{\text{Good}})=0.6500 $$

The remaining attributes are Income and Existing Loan.

Attribute	Weighted Entropy	Information Gain
Income	0.1667	0.4833
Existing Loan	0.4591	0.1909

Therefore, Income is selected.

- High income: all records are Approved.
- Medium income: all records are Approved.

- Low income: one Approved and one Rejected record remain, so split using Existing Loan.
    - Existing Loan = No → Approved.
    - Existing Loan = Yes → Rejected.

9.2 For Credit Score = Poor

This subset contains 6 records: 2 Approved and 4 Rejected.

\[H(S_{\text{Poor}})=0.9183 \]

Attribute	Weighted Entropy	Information Gain
Income	0.6667	0.2516
Existing Loan	0.4591	0.4591

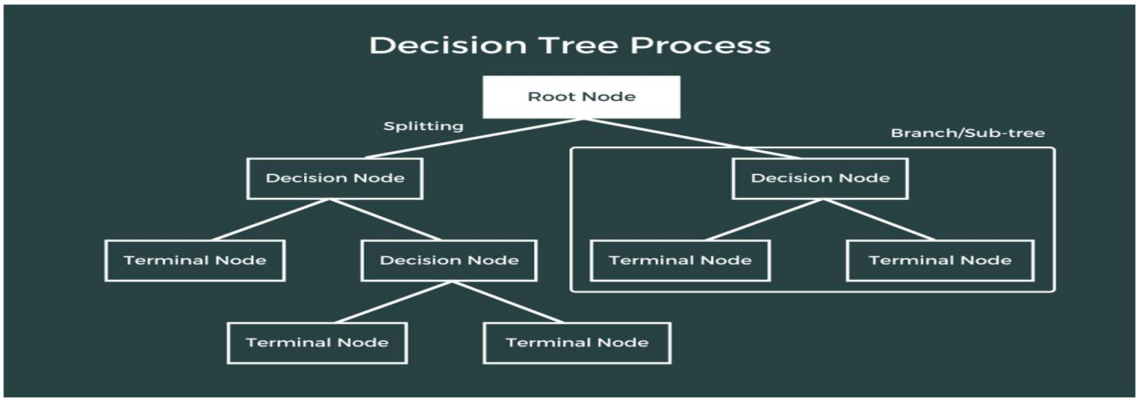
Therefore, Existing Loan is selected.

- Existing Loan = Yes: all records are Rejected.
- Existing Loan = No: the remaining records are divided using Income.
    - High income → Approved.
    - Medium income → Approved.
    - Low income → Rejected.

These calculations follow the ID3 rule of selecting the attribute with the highest Information Gain at each node.

10. Final Decision Tree Diagram:

- The tree starts with Credit Score because it has the highest Information Gain (tied with Existing Loan).
- The Good branch is divided by Income, while the Poor branch is divided by Existing Loan.
- Further splitting is performed only for the subsets that contain both target classes.



Explanation: The tree starts with Credit Score because it has the highest Information Gain (tied with Existing Loan). The Good branch is divided by Income, while the Poor branch is divided by Existing Loan. Further splitting is performed only for the subsets that contain both target classes.

11. Classification of Two New Applicants:

After constructing the decision tree, it can be used to classify applicants whose loan decisions are unknown.

Applicant	Income	Credit Score	Existing Loan	Predicted Decision
B1	High	Poor	Yes	Rejected
B2	Medium	Good	Yes	Approved

Applicant B1

- Income = High
- Credit Score = Poor
- Existing Loan = Yes

Decision path: Credit Score → Poor → Existing Loan → Yes → Rejected.

Therefore, the predicted loan decision is Rejected.

Applicant B2

- Income = Medium

- Credit Score = Good
- Existing Loan = Yes

Decision path: Credit Score → Good → Income → Medium → Approved.

Therefore, the predicted loan decision is Approved.

12. Advantages of Decision Tree Induction:

1. Easy to understand: The tree represents the decision process in a simple visual structure.
2. Interpretable rules: Each path from the root to a leaf explains how a classification is obtained.
3. Handles categorical attributes: ID3 can work with attributes such as income level, credit score category, and loan status.
4. No complex mathematical prediction at the final stage: Once the tree is constructed, new applicants can be classified by following its branches.
5. Useful for decision support: It can help financial organizations understand patterns in historical loan decisions.

13. Limitations:

1. ID3 may overfit the training dataset when the tree becomes too complex.
2. It can be sensitive to changes in the training data.
3. Basic ID3 is primarily designed for categorical attributes; continuous attributes such as exact salary or numerical credit scores need appropriate handling.
4. A tree learned from a small dataset may not generalize well to real applicants.
5. The model reflects patterns in historical decisions and may reproduce bias in those decisions.

14. Applications:

Decision Tree Induction can be used in:

- Banking and loan approval prediction.

- Customer classification and risk assessment.
- Medical diagnosis support.
- Student performance prediction.
- Fraud detection and insurance claim analysis.
- Marketing and customer purchase prediction.

15. Result and Conclusion:

The ID3 algorithm was applied to a sample dataset of 12 loan applicants to construct a decision tree for loan approval prediction. The dataset contained 7 Approved and 5 Rejected records. The initial entropy was calculated as approximately 0.9799.

Information Gain was calculated for Income, Credit Score, and Existing Loan. Credit Score and Existing Loan achieved the highest initial Information Gain of approximately 0.1957. Credit Score was selected as the root attribute, and further splits were selected by recalculating Information Gain within each subset.

The final decision tree successfully classified two new applicants: B1 as Rejected and B2 as Approved. Thus, the experiment demonstrates how decision tree induction converts historical data into interpretable classification rules.